13th September, 2024.

Dr. Kim C. Dickey

Instructor

Department of Chemical and Biomolecular Engineering

North Carolina State University

Dear Dr. Dickey,

Attached is the final report for the experiment "Batch Distillation". The main objective of this experiment was to learn how to operate a batch distillation unit by distilling a mixture of methanol and isopropyl alcohol and generating a product composed of purified methanol, then compare the results of using a higher and lower reflux ratio. It was discovered that using a higher reflux ratio of 1.5 will result in a higher purity product compared to that of 0.75. Data acquired throughout the experiment was analyzed using various Rayleigh analysis techniques, a McCabe Theile analysis, and a Matlab simulation to fully understand the process.

Using a rigorous Rayleigh analysis, we were able to independently check the material balance of both runs and produce plots of the methanol weight fraction of the distillate as a function of time. In conjunction with the various Rayleigh analysis techniques, other calculations were performed, including a McCabe Theiele analysis and a MATLAB simulation. Together, these calculations highlighted the effect that reflux ratio has on the purity of the distillate product.

Using different approaches to analyzing the data collected during the experiment and comparing the results enabled a better understanding of the process itself. This experiment enabled the six of us to visualize the impact and importance that the reflux ratio of the batch distillation has on the distillate concentration.

We appreciate your effort in reviewing our research.

Sincerely yours,

Aaron Dornseif, Robert Hart, Raghu Vansh Mereddy, Elisa Schlesner Alves Jose B

Batch Distillation

Group 10

Jose Maria Barbeito, Robert Hart, Elisa Schlesner Alves, Linda Pomiranceva, Aaron Dornseif,

Raghu Mereddy

Abstract

The Batch Distillation experiment analyzes the distillation of two liquids. In this experiment, methanol and isopropanol are separated in a distillation column, using three trays and a condensing system. Since methanol is more volatile, it was expected to evaporate more easily and be found in higher concentrations in the distillate compared to the bottoms. Unlike continuous distillation, batch distillation is a transient process used to produce high-value products with less feed. The apparatus used in this experiment included a condenser, separation trays, a sampling valve, a reboiler, and a control panel for setting parameters like the reflux ratio.

The mixture of isopropanol and methanol was raised to a temperature of 71℃, at which point the liquid began to boil and vapor was collected and condensed, filling bubble trays. Over 35 min, the distillate was collected in nineteen containers and analyzed using a refractometer to determine methanol weight%. After the first run was completed with a reflux ratio of 1.5, a second run was performed with a reflux ratio of 0.75. The results of both were analyzed using various methods.

The first, a rigorous Rayleigh analysis, found that with a reflux ratio of 1.5, the methanol mole fraction in the distillate decreased from 0.81 to 0.73 after 35 minutes. The methanol mole fraction in the still itself decreased from 0.57 to 0.51. The calculated and experimental values for trial one were within 10%, being 0.215 (experimental) compared to 0.219 (calculated).

Secondary analysis, using McCabe-Thiele and MATLAB simulation techniques, highlighted how lower reflux ratios yielded more distillate with lower purity, while higher reflux ratios improved separation at the cost of overall yield. Additionally, these calculations tended to undervalue the importance of the reflux ratio, when compared to experimental values.

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Introduction

Batch distillation is a handy tool for the separation process industry. Batch distillation is crucial for certain chemical engineering disciplines, such as pharmaceuticals, biochemicals, and specialty chemical production. Batch distillation allows chemical engineers to have acute control over their process, making it possible to create more difficult and high-value products [1]. Additionally, batch distillation requires less control equipment, allows for more flexibility, is more easily cleaned, and the feedstock that can be used is more diverse [2]. To ensure that aspiring chemical engineers understand how batch reactors function, the following experiment was created.

In this experiment, participants will distill a mixture of methanol and isopropanol to produce a product of purified methanol. By performing this experiment, participants can observe the separation characteristics of a methanol-isopropanol mixture. Additionally, this experiment aims to teach participants how to operate a batch distillation unit, gather accurate data, and analyze their findings.

The experiment will be performed by running two separate batch distillations of methanol and isopropanol, each with different reflux ratios. For each experiment, participants will gather the temperature, mass, and refractive index following a predetermined timetable. With these values, the participants will be able to calculate values such as the batch mass fraction, total mass of distillate, and the vapor mass fraction. Later on, these values will be used to analyze and compare the effect of changing a batch distillation's reflux ratio on distillate concentration [1].

Theory

The Rayleigh equation, introduced below, is the basis to start the analysis of most distillation processes [1].

$$\int_{N^L_o}^{N^L(t)} dN^L(t)/N^L(t) = \ln[N^L(t)/N^L_o] = \int_{x_{io}}^{x_i(t)} dx_i(t)/[y_i(t) - x_i(t)] \tag{1}$$

In this experiment, participants will analyze and investigate the data using five different approaches, and further compare the results among them. The first one is called the Rigorous Rayleigh Analysis, in which the left-hand side of Equation 1 is calculated based on data gathered at the beginning and end of the experiment. The methanol mole fraction of what remains in the still at time t can be found by the evaluation of a thorough and precise component balance on both methanol and isopropanol after each condensate sample is reserved [1], shown in equation 2 below.

$$x_{M\ n}(t) = \frac{\frac{1}{MW_M}[M_o w_{M\,o} - \sum_{i=0}^{n(t)} m_i w_{M\ i}]}{\frac{1}{MW_M}[M_o w_{M\,o} - \sum_{i=0}^{n(t)} m_i w_{M\ i}] + \frac{1}{MW_I}[M_o(1 - w_{M\,o}) - \sum_{i=0}^{n(t)} m_i(1 - w_{M\ i})]} \tag{2}$$

The plot of $\frac{1}{yM(t) - xM(t)}$ versus methanol mole fraction, $x_M(t)$, along with the trapezoidal or Simpson's rule allows for the evaluation of the right-hand side. Consistency between the two sides is shown if the calculated results are within 10% of each other [1].

The second approach to determine the right-hand side of equation 1 is an analysis based on a one-stage equilibrium assumption via given VLE data. The plot of y_M vs x_M at equilibrium is used to establish a sequence of $x_M(t)$ values based on the instantaneous distillate samples. This approach is performed similarly to the rigorous Rayleigh analysis; however, the evaluation of methodical and meticulous mole balances on the mixture's components is not necessary.

The third option is to approximate the Rayleigh analysis based on both one-stage equilibrium assumption and constant relative volatility. Relative volatility is commonly used in distillation processes, and its calculation is based on how much the vapor pressure of the more volatile component varies from the less volatile component in a liquid mixture [3]. The higher the resultant vapor pressure is, the easier it is to separate one component from the other. The analysis of the right-hand side of the equation is given by

$$\frac{1}{(\alpha_{MI}-1)}\ln\frac{x_{Mf}(1-x_{Mo})}{x_{Mo}(1-x_{Mf})}+\ln\frac{1-x_{Mo}}{1-x_{Mf}} \tag{3}$$

in which the α_{MI}, assumed to be constant, is the average of the relative volatilities calculated from the VLE data in the composition range. Agreement between the left and right-hand sides of equation 1 will demonstrate if the data collected throughout the experiment is consistent and determine if this approach is valid or not.

$$y_{M(n+1)}=\frac{RR}{RR+1}x_{M(n)}+\frac{1}{RR+1}x_M^{\ D}=\frac{RR}{RR+1}x_{M(n)}+\frac{1}{RR+1}y_M^{\ D} \tag{4}$$

For the fourth alternative to calculate the numerical value of the right-hand side of equation 1, the McCabe-Thiele point-by-point analysis assumes a multi-stage equilibrium and a pseudo-steady state. This analysis is to be performed on two different trials in each of the runs using Equation 4. The known reflux ratio and the measured composition of the chosen samples allow for the construction of a McCabe-Thiele plot, which further aids in determining the composition of the liquid that remains in the still when the instantaneous distillate is collected. A comparison between the composition value that was visually determined using the McCabe-Thiele analysis and Equation 2 is to be made to show how close the values are.

Finally, participants conducted a MATLAB simulation of the batch distillation process for the methanol-isopropanol mixture, incorporating Rayleigh's equation under the assumption of one-stage equilibrium. The simulation utilized Rayleigh's equation to describe the relationship between the number of moles and composition as the distillation progressed. Different sets of differential equations were applied before and after the boiling point, reflecting changes in temperature, total mole number, and methanol mole fraction during the process. The van Laar model was used to account for non-ideal behavior in the mixture through activity coefficients. Initial conditions and experimental data from both trials were incorporated into the simulation to compare system behavior. While the reflux ratio was not accounted for in the MATLAB model, the results provided insight into how varying reflux ratios could influence the mole fraction and purity of the components during distillation.

Experimental

The experiment performed was a batch distillation experiment, with two trials. The experiment materials included pipettes, nineteen jars with lids to collect distillate, one large flask to collect entire stills contents, and one flask to collect the cumulative distillate during the experiment. A scale, magnetic stir bar, stopwatch, refractometer, and distillation apparatus were machines used in the experiment. The required chemicals were methanol and isopropanol. The distillation apparatus had three trays present with a condenser at the top of the column and a reflux switch present within the column. The column as well as the still should be constructed of glass. The heating elements and other metal components should be constructed of stainless steel. Within the control panel, there should be a power switch, temperature control switch, reflux switch, and variac dial for voltage. On the control panel, there should also be readings for voltage, current, and temperature.

Preliminary steps which were completed before the powering of the heater are as follows. First, the experimenters familiarized themselves with the apparatus and control panel so that no damage came to the apparatus. For optimal operation of the experiment the group, of which there were six members, assigned each member a task to maintain during the experiment. There was an individual who collected the samples and switched containers as required, an individual timed the experiment and instructed the sample collector when to swap containers, a mass recorder weighed each sample, a refractive index measurer who used the refractometer, and an individual to record the data that is collected during the experiment. After tasks were assigned the refractive index measurer calibrated the refractometer. To calibrate the refractometer the following steps were completed. A member turned on the refractometer and added a few drops of pure isopropanol until the gray sensor was covered then wiped with a kimwipe. Next, the "25°Cal" button was pressed and the member waited for the temperature to stabilize, then allowed the refractometer to automatically measure the refractive index of air. When prompted to add water a few drops of pure water were added then the lid was closed and the member pressed "OK." When prompted to "Drain cell," the member wiped the sensor with a Kimwipe and then, pressed "OK" when prompted to "Rinse cell with water." When prompted to "Rinse cell with acetone" the member added a few drops of isopropyl and pressed "OK." When prompted to "dry cell" the member used a Khemwipe to clean the sensor and closed the lid, then pressed "OK."

Once the calibration passed, the group continued to measure the refractive indexes of 0, 25, 50, 75, and 100 wt% methanol in isopropanol samples to create a calibration curve between refractive indexes and wt% methanol. If the calibration fails, repeat the calibration steps. To measure the refractive index of a sample a member of the group pressed the “RI°25” button then at the “Add sample” prompt added a few drops of the sample and pressed “OK.” The member dried the sensor once prompted to “Drain cell”, shut the lid, and pressed “OK.” The member recorded the displayed refractive index and pressed “OK” then added a few drops of isopropanol to the sensor and wiped it dry with Kimwipes to prepare it for the next sample. The group recorded the weight of each jar or flask to be used in the experiment and labeled the jars based on the sampling times. Recommended sampling times are as follows 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20, 23, 26, 29, 32, 35 minutes. The group mixed the solution to be used in this experiment consisting of approximately 1600 mL of methanol and 2400 mL of isopropyl alcohol mixed thoroughly. The initial mass of this mixed solution was measured as well as its refractive index. The cold water supply to the condenser was turned on. The group adjusted the reflux switch to a reflux ratio of either 1.5 or 0.75. In our experiment, a reflux ratio of 1.5 was used first, followed by a reflux ratio of 0.75.

To begin the experimental run, the power cable was plugged into the wall receptacle and the variac was set to below 220V, in this experiment the voltage was set to 152V for the entire duration. The group ensured the sample jars and the cumulative distillate flasks were easily accessible at the start of the experiment. The time when the distillate first drips from the sampling stem was noted and then the timing of the experiment began. Sample jars were removed and replaced once the experiment runtime had reached the labeled time on the specified jar. For each jar, once it had collected its sample, the sample weight was measured then a sample was taken to be used in the refractometer. Once measurements of a sample were concluded the continents were poured into the cumulative distillate flask. Once a runtime of 35 minutes had been reached, heater power and reflux control were turned off. As the still cooled, the cumulative distillate was mixed then measured of its mass refractive index. The group drained the remaining solution from the still and measured its mass and refractive index. Next, the cumulative distillate and the drained solution were measured for the resulting solution's mass and refractive index. A second trial using the alternative reflux ratio was performed, following the same experimental steps.

Results and Discussion

Analysis of the data gathered is separated into five sections, a rigorous Rayleigh analysis, an approximate Rayleigh analysis using relative volatility, an approximate Rayleigh analysis via vapor-liquid equilibrium (VLE), a McCabe Thiele analysis, and a Matlab simulation assuming one-stage equilibrium & Van Laar activity coefficients.

Part A: Rigorous Raleigh Analysis

The left-hand side of Equation 1 can be approximated by using data collected at the beginning of the run with the following equation:

$$\ln\frac{N^L{}_f}{N^L{}_o} = \ln\frac{M_f[\frac{w_{M\,f}}{MW_M}+\frac{1-w_{M\,f}}{MW_I}]}{M_o[\frac{w_{M\,o}}{MW_M}+\frac{1-w_{M\,o}}{MW_I}]} \tag{4}$$

where $N^L{}_o$ and $N^L{}_f$ are the mole numbers of the solution in the still at the start and end of the run, respectively, W_o and W_f are the weights of the solution in the still at the start and end of the run, respectively, MW_M and MW_I are the molecular weights of methanol and isopropanol, respectively, and w_{Mo} and w_{Mo} are the methanol weight fractions of the solution in the still at the start and end of the run, respectively. Using this equation, the mass of each sample of distillate collected, and the refractive index of each sample (to calculate the methanol weight fraction using our calibration curve), we calculated $\ln(N^L_f / N^L_o)$ to be approximately 0.219.

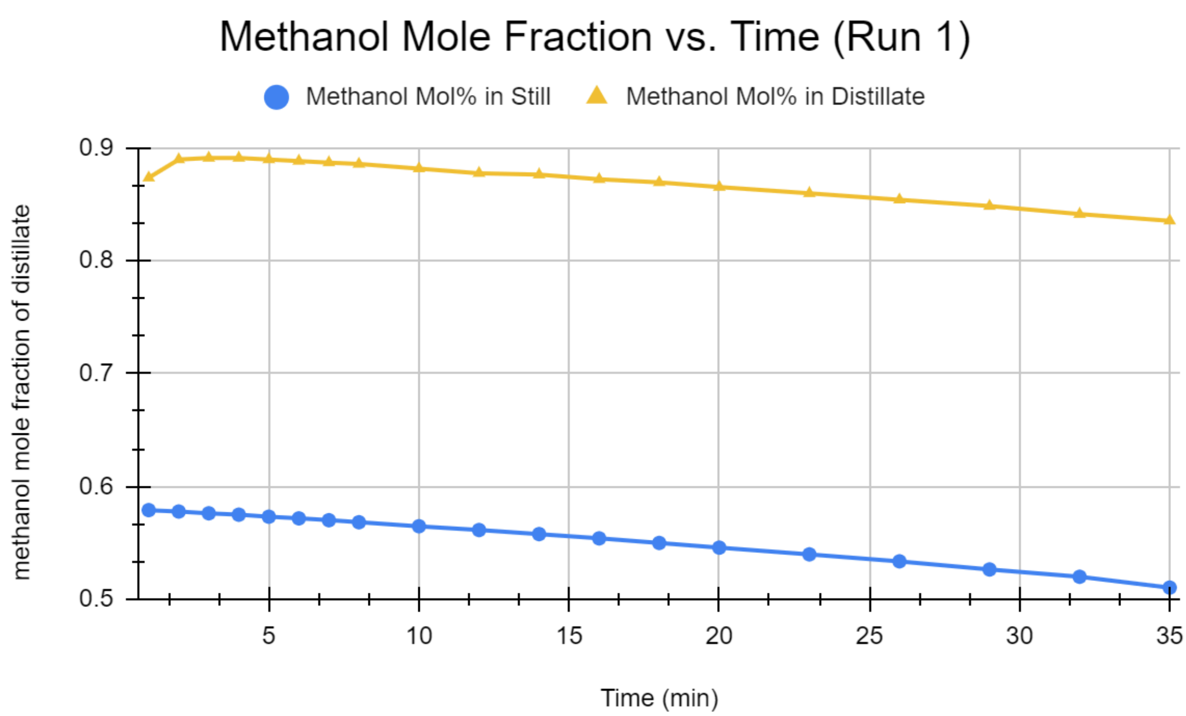

Figure 1. Methanol Mol Fraction vs. Time (Run 1).

Using this data, we could plot the methanol mol fraction in the distillate and the still over time, as shown in Figure 1. At the beginning of the experiment, the first samples contain relatively low amounts of methanol, due to the need for the distillation column trays needing to be primed. After the first couple of samples were collected, the maximum methanol mol fraction was achieved, at 0.89. Over time, this fraction dropped slowly to a final value of 0.83. The methanol percent in the still can also be seen, starting at a fraction of 0.57 and ending at a fraction of 0.51.

Using this data we can calculate the right side of Equation 1 by plotting the data (shown below in Figure 2). Then, using the trapezoidal rule, we can estimate the area underneath the curve from x_{Mo} to x_{Mf}.

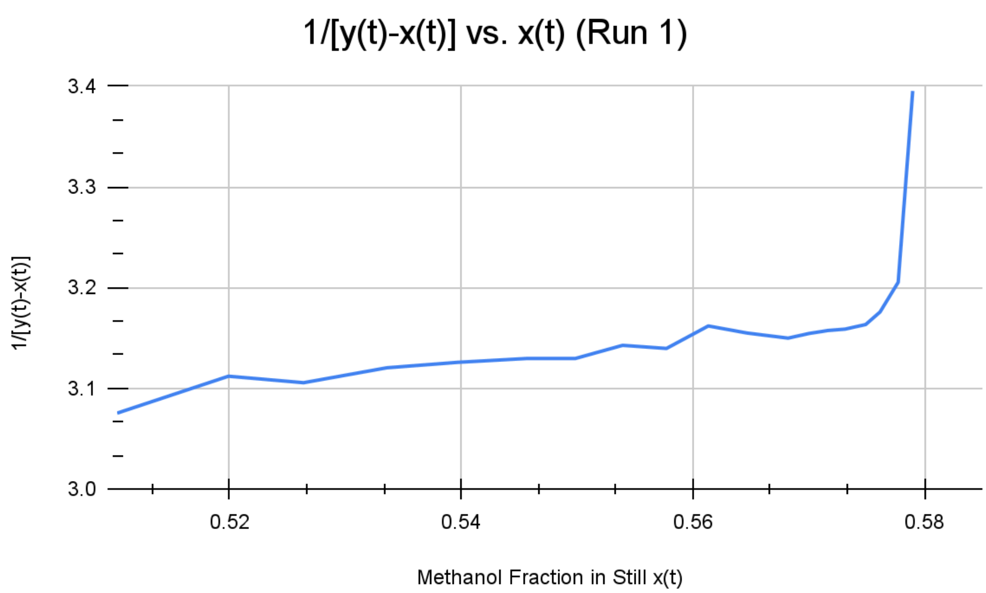

Figure 2. 1/[y(t)-x(t)] vs. x(t) (Run 1).

After using the trapezoidal rule, the area under the curve was calculated to be 0.215. Since this value is within the ± 10% expected, it can be concluded that the data is consistent between the two analytical methods. Analyzing the second distillation run using the same methods, Figure 3 and Figure 4 were made, showing the methanol mol fraction over time and 1/[y(t)-x(t)] vs. x(t). In addition, using Equation 4, $\ln(N^L{}_f / N^L{}_o)$ was calculated to be approximately 0.747, while the area under the curve of Figure 4 was calculated to be 0.616 using the trapezoidal rule. Potential sources of error include human malpractice, inconsistent measuring techniques, and loss of material over the run due to evaporation.

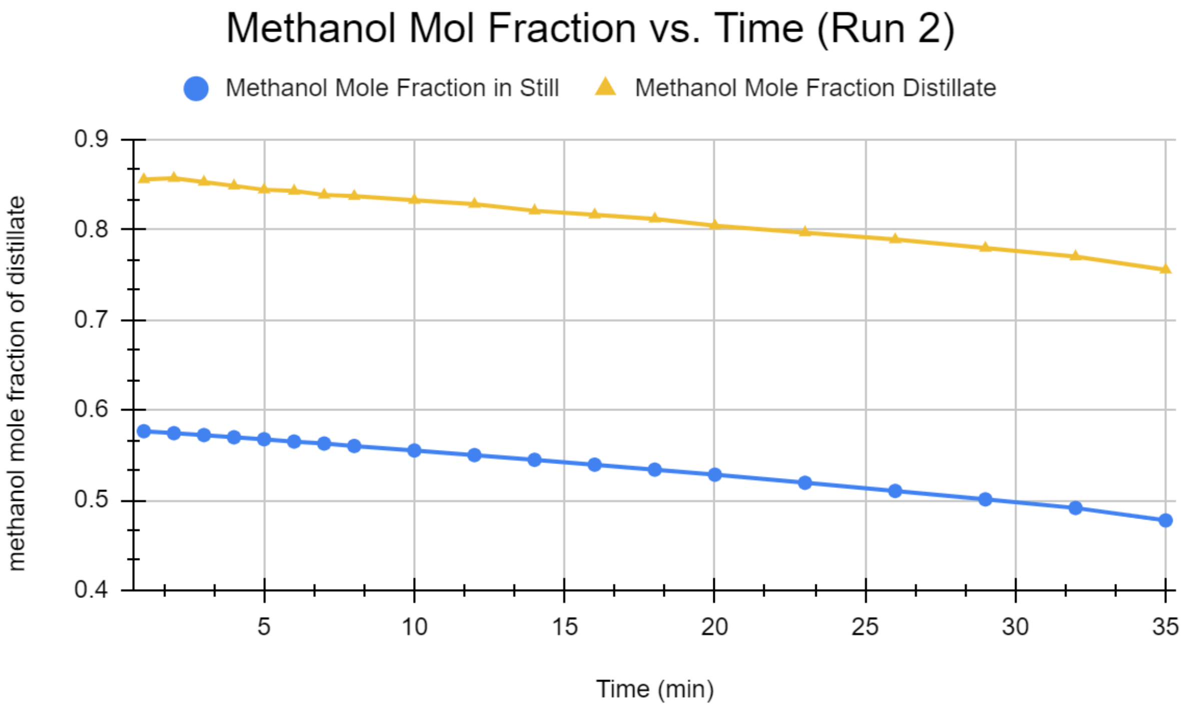

Figure 3: Methanol Mol Fraction vs Time (Run 2).

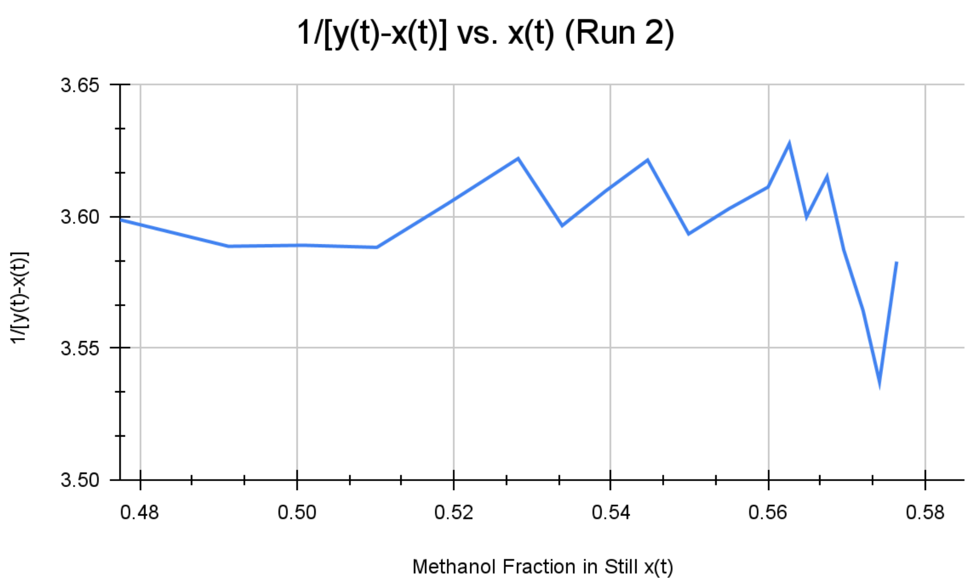

Figure 4. 1/[y(t)-x(t)] vs. x(t) (Run 2).

In conclusion, calculated and experimental values only matched each other for the first trial, while the second trial had a larger amount of discrepancy. Despite this, the analysis was crucial in understanding the methanol mole fraction in the distillate and in the still over time for both runs.

Part B: Approximate Rayleigh Analysis using only VLE data

A one-stage equilibrium assumption was used in conjunction with vapor-liquid equilibrium data to analyze the data collected from the experiment. The steps of this analysis were as follows. First, the mass fraction collected was converted to its mole fraction using the molecular weight of the chemicals in the solution. The mole fraction of methanol in the distillate can be set equal to the vapor fraction of methanol in the still. Using the Vapor-Liquid Equilibrium correlation, the mole fraction of methanol in the liquid phase of the still can be determined from the liquid fraction of methanol in the distillate. This relationship can be determined either graphically or mathematically by finding the temperature of the solution at a given y_m and finding the x_m value where it equals the same temperature.

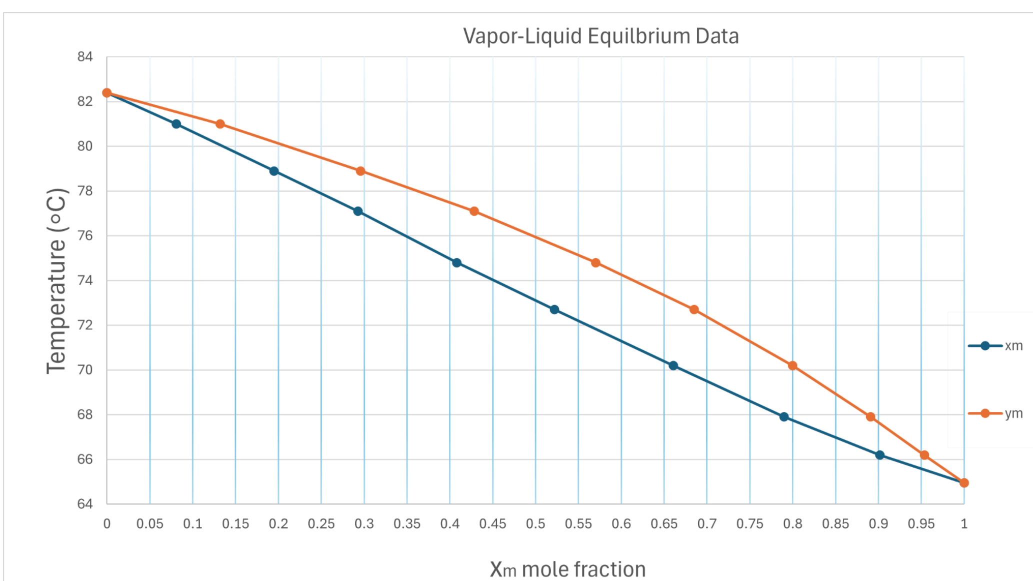

Figure 5: Graphical representation of VLE data.

After determining the vapor and liquid fractions of methanol these values were used to evaluate the right side of equation 6 and calculated using the trapezoidal rule. The value was calculated to be 0.572 for the first trial. Comparing this value to the left hand side of equation 6 shows a 89% difference between the values, indicating one one-stage equilibrium is not a good assumption for this batch distillation. Plotting $1/(y_m-x_m)$ vs x_m correlation, shows as x_m increases $1/(y_m-x_m)$ increases, having an exponential correlation.

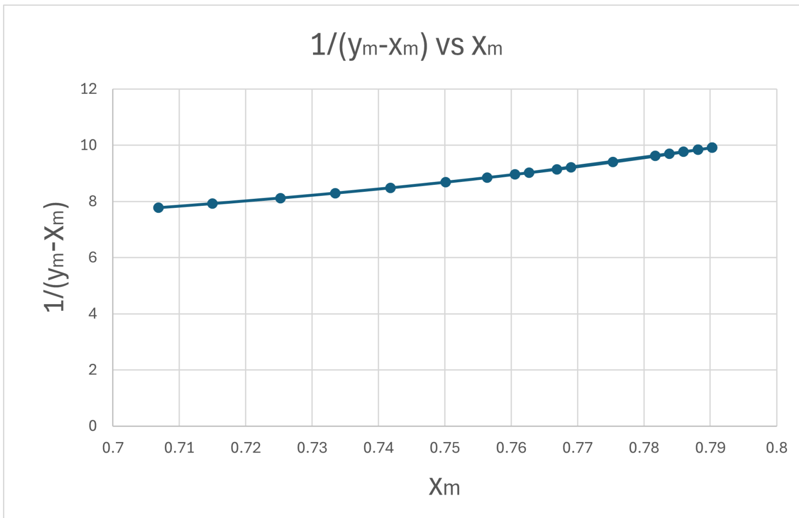

Figure 6: 1/[y(t)-x(t)] vs. x(t) (Run 1) using One-stage equilibrium assumption and VLE.

The second trial shows a similar exponential relationship between $1/(y_m-x_m)$ and x_m. When evaluating the right side of equation six using the trapezoidal rule the value is found to be 0.967 this value is closer to the calculated left side value for equation 6, but still varies substantially. There is a 25.7% difference between the values indicating a one-stage equilibrium may not be a reasonable assumption for this scenario.

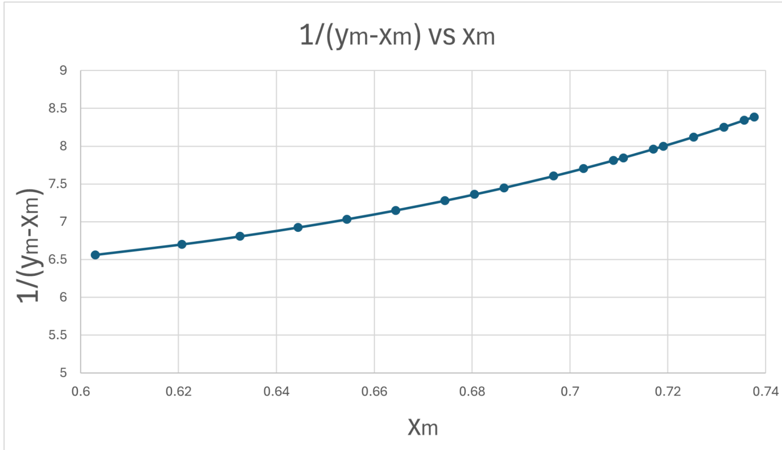

Figure 7: 1/[y(t)-x(t)] vs. x(t) (Run 2) using One-stage equilibrium assumption and VLE.

Part C: Approximate Rayleigh Analysis via Vapor-Liquid Equilibrium and Relative Volatility

In conjunction with a one-stage equilibrium assumption, constant relative volatility was assumed to estimate results for the right-hand side of equation 1 using this approach. Being a binary mixture, it was possible to infer that $y_I = 1\text{-}y_M$ and $x_I = 1\text{-}x_M$ and further use Equation 3 to establish results for the calculations. The average relative volatility was found to be 2.950 for trial 1 and 2.082 for trial 2, and was taken as the mean of the relative volatilities calculated for the interest range of the compositions of the components. Plugging this value and substituting the known values for the compositions in Equation 3, the calculated value for this approach was 0.665 for trial 1, showing a relative agreement with the left-hand side. However, for trial 2, the calculated value was 1.93, showing a non-neglectable deviation from the left-hand side.

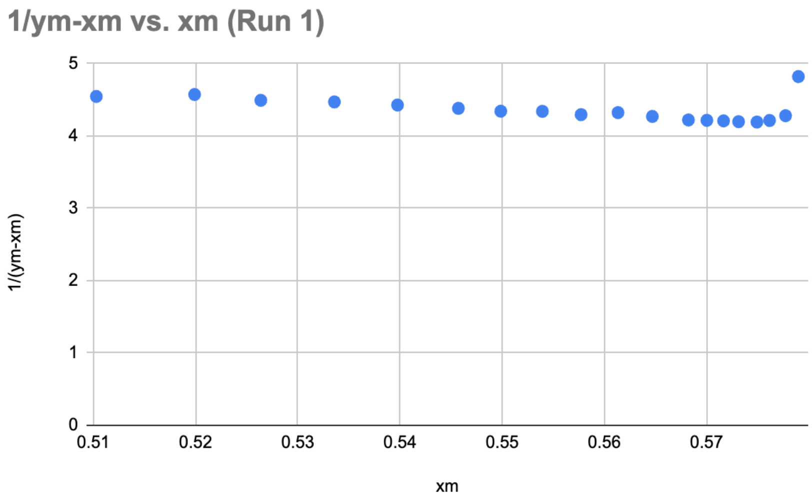

Figure 8: 1/[y(t)-x(t)] vs. x(t) (Run 2) using relative volatilities.

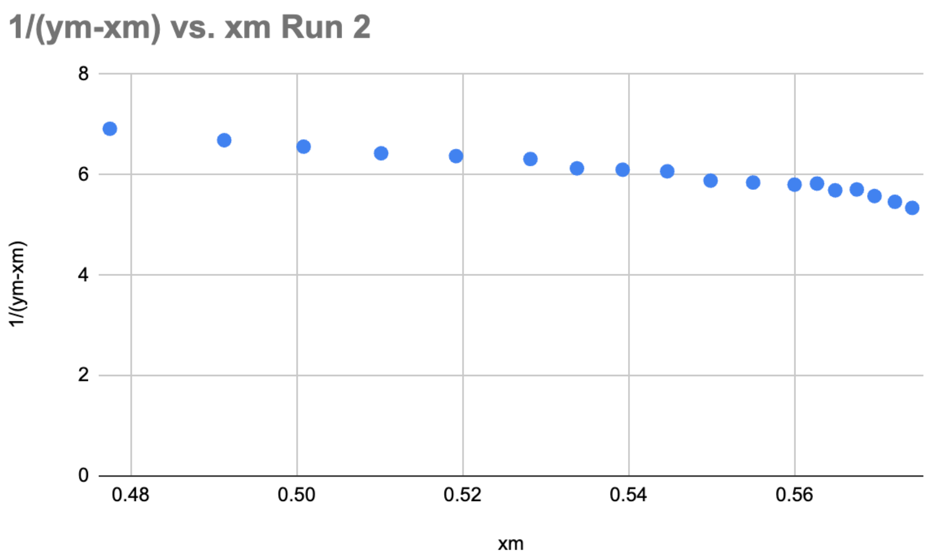

Figure 9: 1/[y(t)-x(t)] vs. x(t) (Run 2) using relative volatilities.

Part D: McCabe Thiele Analysis

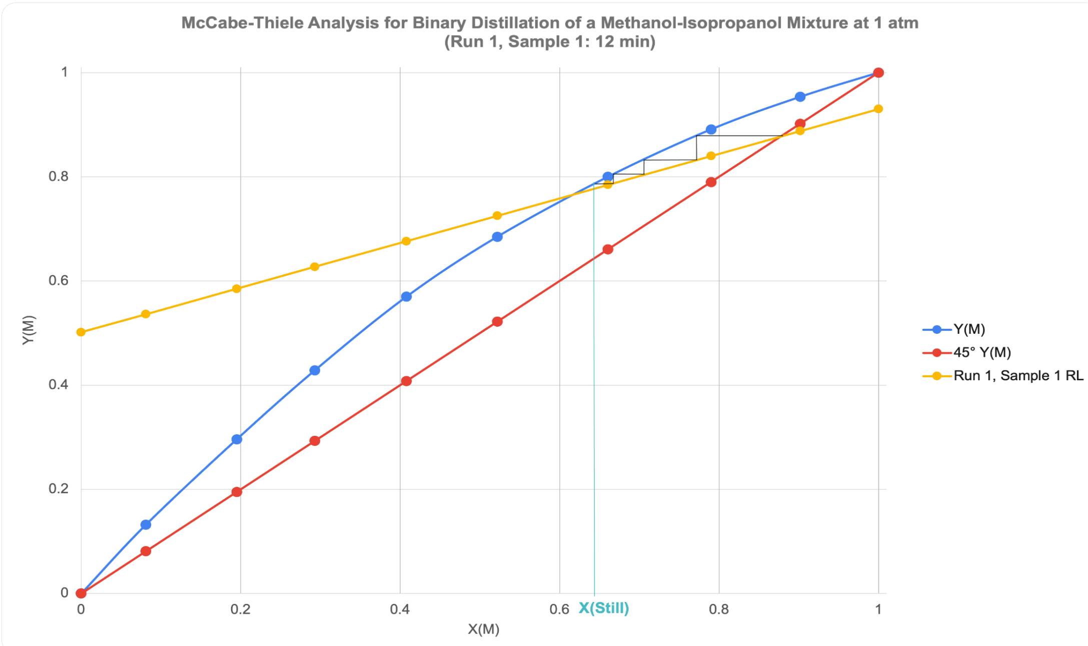

Figure 10. McCabe-Thiele operating line for a reflux ratio of 1.5.

Figure 11. McCabe-Thiele operating line for a reflux ratio of 1.5.

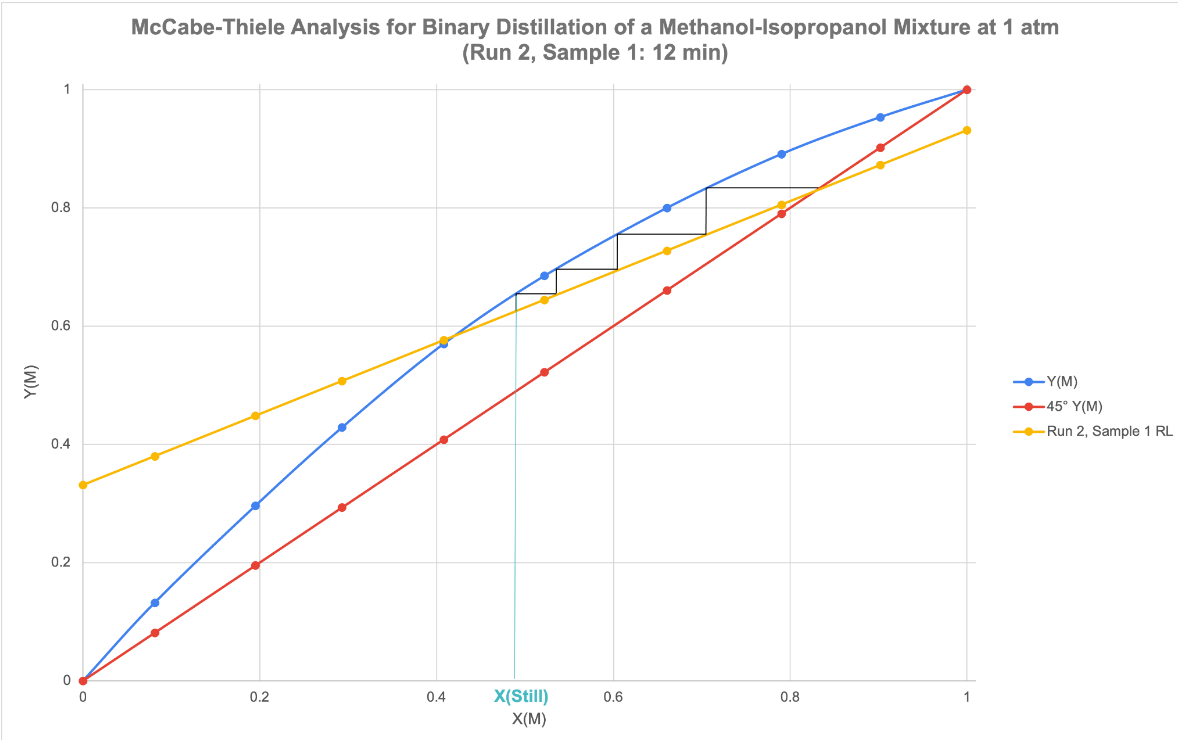

Figure 12. McCabe-Thiele operating line for a reflux ratio of 0.75.

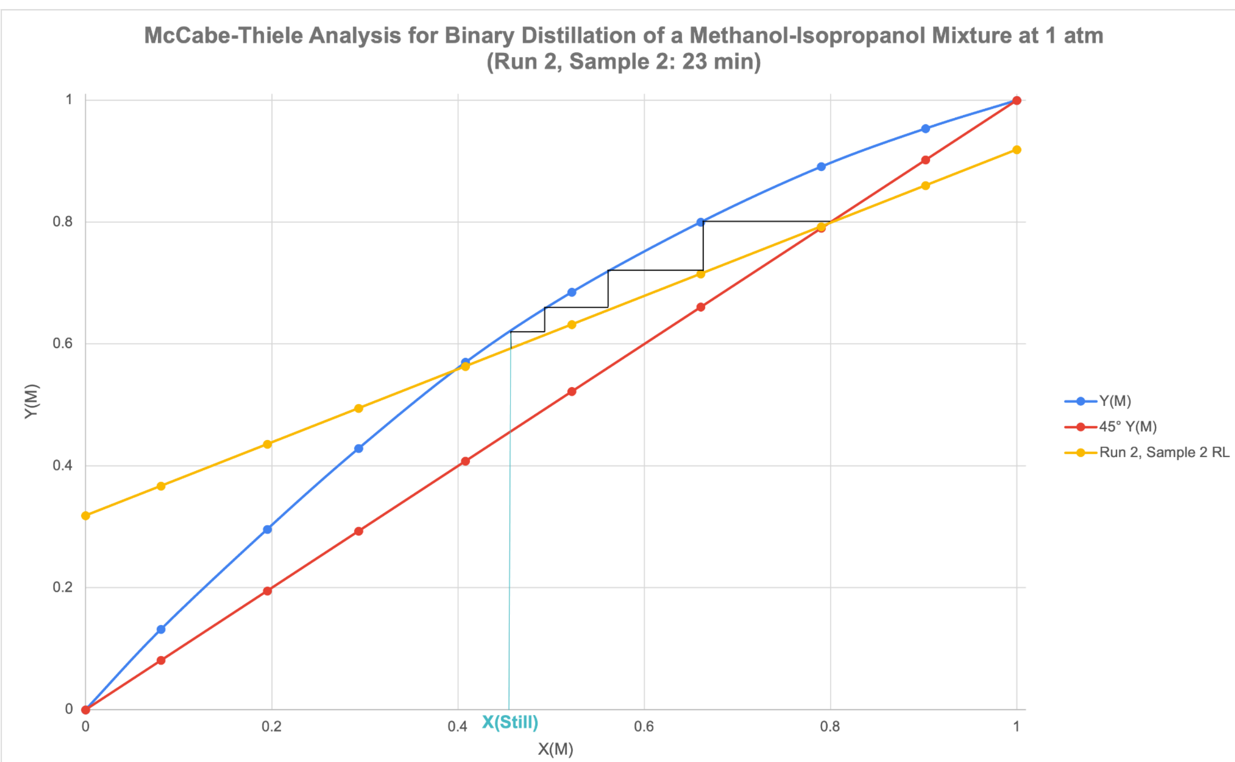

Figure 13. McCabe-Thiele operating line for a reflux ratio of 0.75.

A McCabe-Thiele analysis was conducted for two points in each trial of the batch distillation process assuming a multi-stage equilibrium, equimolal overflow, and pseudo steady state. Based on these assumptions, the operating lines for the enriching section of the distillation column, which extends from the top stage to just above the feed stage, could be graphed using Equation (4). The four equilibrium stages in the experimental setup were easily stepped off between the methanol vapor-liquid equilibrium curve and operating line for the enriching section of the distillation column. The composition of the solution remaining in the still at the instant the distillate sample was taken could then be graphically determined as the methanol mole fraction of the fourth equilibrium stage.

For trials 1 and 2, with respective reflux ratios of 1.5 and 0.75, instantaneous samples of distillate were analyzed at 12 and 23 minutes. For trial 1, the methanol mole fraction of the solutions remaining in the still at 12 and 23 minutes were approximately 0.64 and 0.62, respectively (Figure 8-9). Trial 2 displayed methanol mole fractions equal to 0.48 and 0.46 at 12 and 23 minutes. It is known that a higher reflux ratio improves the separation of components by increasing the amount of contact between the two phases in the column. Thus, higher-purity distillate should equate to a lower concentration of methanol in the still as observed with the graphically-obtained McCabe-Thiele values.

When comparing the graphically determined composition values of the solution in the still with those calculated in the rigorous Rayleigh analysis, slight variations were observed. The methanol mole fractions obtained through the rigorous Rayleigh analysis for trial 1 were about 0.56 (12 minutes) and 0.54 (23 minutes). Calculations performed for trial 2 yielded values of 0.55 (12 minutes) and 0.52 (23 minutes). This slight discrepancy in values could be attributed to the possibility that the assumptions inherent in a McCabe-Thiele analysis were not completely valid in this experiment.

Part E: Matlab Simulation Assuming One-Stage Equilibrium & Van Laar Activity Coefficients

In this study, a MATLAB simulation was employed to model the batch distillation process of a methanol-isopropanol mixture, following the Rayleigh distillation assumption. The simulation utilized differential equations to describe the system's key parameters: temperature (T), total mole number (N), and the mole fraction of methanol (X_m). One-stage equilibrium was assumed, and the Van Laar model was applied to account for the non-ideal behavior of the mixture. These equations were solved numerically in MATLAB, with distinct equations applied before and after the onset of boiling.

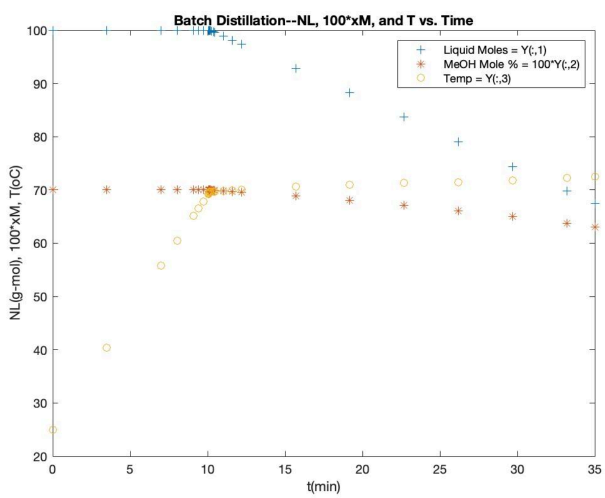

Figure 14: Matlab Graph of Initial Conditions

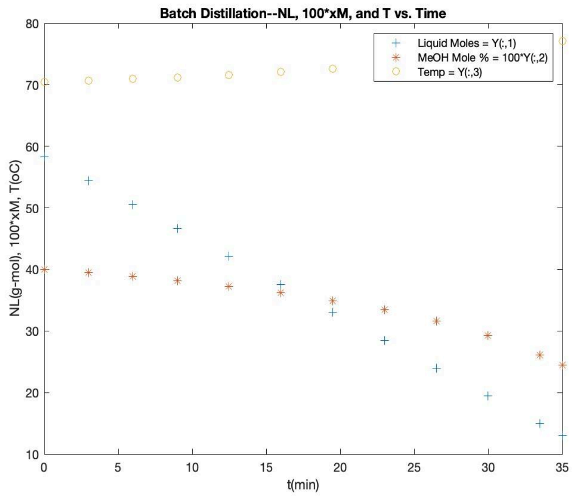

Figure 15: Matlab Graph of Trial 1 at Reflux Ratio 0.75

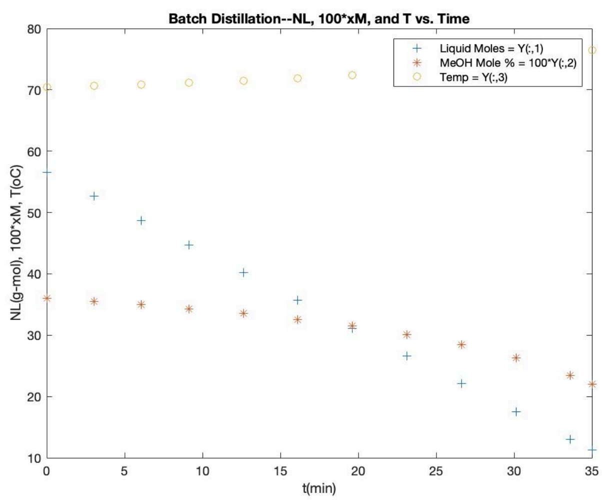

Figure 16: Matlab Graph of Trial 2 at Reflux Ratio 1.5

For the simulation, initial conditions of N_o = 100 g-mol, X_{mo} = 0.70, and T = 25°C were used for the provided MATLAB data as can be seen in Figure 14. Experimental data from two trials were then compared: Trial 1 (N_o = 58.33 g-mol, X_{mo} = 0.36, and T = 70.4°C) and Trial 2 (N_o = 56.57 g-mol, X_{mo} = 0.36, and T = 70.7°C). The resulting figures demonstrated the system's behavior during the distillation process. The simulation results showed that Trial 1, with a reflux ratio of 0.75, yielded a lower methanol mole fraction at the end of the process compared to Trial

2, which had a reflux ratio of 1.5. This outcome aligns with the expected trend of a lower reflux ratio producing more yield but with less purity, while a higher reflux ratio increases purity at the expense of yield. However, because the MATLAB simulation did not account for the reflux ratio, it predicted a faster increase in temperature and a steeper decline in methanol mole fraction than what was observed experimentally.

In Trial 1 of the experiment, the MATLAB simulation was compared to the batch distillation process for a methanol-isopropanol mixture with initial conditions of No = 58.33 g-mol, Xmo = 0.36, and T = 70.4°C. The reflux ratio for this trial was 0.75, meaning that less vapor was condensed and returned to the distillation column, resulting in a greater overall yield but lower product purity compared to a higher reflux ratio.

In Trial 2, the MATLAB simulation was again used to analyze the batch distillation of the methanol-isopropanol mixture, but with different initial conditions: N_o = 56.57 g-mol, X_{mo} = 0.36, and T = 70.7°C. This trial employed a higher reflux ratio of 1.5, meaning that a greater portion of the condensed vapor was returned to the distillation column, leading to a higher product purity but lower yield compared to Trial 1.

As with Trial 1, the MATLAB simulation, which assumed one-stage equilibrium and excluded the effect of reflux ratio, predicted a faster temperature increase and a quicker decline in the methanol mole fraction compared to the experimental data. In reality, the higher reflux ratio in Trial 2 slowed down the distillation process, resulting in a more gradual decrease in methanol mole fraction and a slower temperature rise. The simulation's exclusion of this reflux effect led to a less accurate representation of the system, as the experimental data showed a much more controlled separation process.

Although the MATLAB simulation produced precise predictions based on the assumption of single-stage equilibrium and the van Laar activity coefficient model, it failed to replicate the experimental conditions fully. The model did not include key factors, such as the impact of the reflux ratio and heat losses to the surroundings, leading to discrepancies between the simulation and experimental results. These limitations in the MATLAB simulation explain why it projected a faster temperature increase and a more rapid decrease in methanol mole fraction compared to the experimental trials.

Conclusion

By performing rigorous data analysis of the batch distillation, the effect of increasing the reflux ratio was determined to increase the composition of methanol in the distillate. The experiment apparatus used to determine the conclusion consists of a large glass container with a heating coil at the bottom, with three distillation shelves, and an eventual distillate outlet at the top.

Two distillation trials were conducted with initial solution concentrations of 57 wt% methanol, using reflux ratios of 0.75 and 1.5. During each trial, temperature, mass, and refractive index data were collected to track the distillate composition over time.

The Rayleigh equation was used for the first round of data analysis, with both rigorous and approximate approaches applied. In the first trial, the methanol mole fraction in the distillate peaked at 0.87 and decreased to 0.83 after 35 minutes. The methanol mole fraction in the still decreased from 0.57 to 0.51. In the second trial, similar trends were observed, though the lower reflux ratio led to slower distillation and a lower final product purity.

Further analysis using McCabe-Thiele and MATLAB simulation techniques demonstrated that lower reflux ratios yielded more distillate with lower purity, while higher reflux ratios improved separation at the cost of overall yield. Quantitative comparisons between experimental and simulated data revealed that the simulation's predictions underestimated the impact of reflux, suggesting improvements in future model accuracy.

By performing this batch distillation experiment, our group learned how to correctly operate and perform tests using a batch distillation chamber. Additionally, we learned the importance of the reflux ratio and how it is often underestimated by calculations.

Recommendations

One recommendation is that the method for collecting the sample be revised. Using a clamp that needs to be tightened and untightened makes timing sample transfer difficult. A proposed solution could be to have a track-type design. An example could be a stand with rails on two sides and an open end on the other, where a jar rests under the distillate exit stream and is secured from falling by the rails. When it is time, push the next jar into the current one allowing for a quick and easy transfer. In addition, better instruction for the analysis would have been helpful, at the minimum a range of expected values would have saved the team a lot of time. Much of the analysis required us to "plug numbers into this equation and see what you get", with little explanation as to what these numbers meant.

Nomenclature

Variable	**Definition**
A_I, B_I, C_I	Antoine coefficients for isopropanol
A_M, B_M, C_M	Antoine coefficients for propanol
$\underline{C}_{PI}$, $\underline{C}_{PM}$	Molar heat capacities of isopropanol and methanol
$\Delta\underline{H}_{vap}$ I, $\Delta\underline{H}_{vap}$ M	Molar heat of vaporization for isopropanol and methanol
K_I, K_M	Kay factors for isopropanol and methanol
m_i	The weight of the i^{th} instantaneous distillate sample withdrawn from the still
M_o, M_f	Weights of the solution in the still at the start and end of the run
$M^{TotD}(t)$	Weight of the cumulative distillate collected up till time t, including the n^{th} instantaneous distillate sample
MW_I, MW_P	Molecular weights of isopropanol and methanol
N^L_o, N^L_f	Total moles of solution in the still at the start and end of the run
$N^L(t)$	Total moles of solution left in the still at time t
P^{sat}	Saturated vapor pressure
Q_o	A specified (constant) rate of heat supplied to the still in the simulation model
R	Reflux ratio
T_o	The initial temperature of the solution in the still
$W_{M\,o}$, $W_{M\,f}$	Methanol weight fractions of solution in the still at the start and end of the run
$W_{M\,i}$	The methanol weight fraction of the i^{th} instantaneous distillate sample

$W_M{}^{TotD}$ (t)	The methanol weight fraction of the cumulative distillate up to time t
x_I	Isopropanol mole fraction of solution in the still
$X_{M\,o}$, $X_{M\,f}$	Methanol mole fractions of solution in the still at the start and end of the run
$X_{M\,n}$ (t) or X_M(t)	Methanol mole fraction of the solution remaining in the still after the n^{th} instantaneous condensate sample is drawn at time t
$X_M{}^D$(t)	Methanol mole fraction of the instantaneous distillate sample drawn at time t
X_M (n)	Methanol mole fraction of the liquid stream leaving the n^{th} stage and falling to the $(n+1)^{th}$ stage directly below it
y_I, y_M	Vapor mole fractions of isopropanol and methanol in the still
Y_M (n+1)	Methanol mole fraction of the vapor stream leaving the $(n+1)^{th}$ stage (counting from the top) and rising to the n^{th} stage directly above it
α	One of the two constants in the van Laar model for activity coefficients
α_{MI}	Relative volatility for the methanol-isopropanol pair
β	The second constant in the van Laar model for activity coefficients
γ_I, γ_M	Activity coefficients of isopropanol and methanol

References

[1] M. Cooper, P. Lim, K. Dickey, and K. Dickey, "Batch Distillation Experiment" CHE 331 Course Handout, North Carolina State University, Raleigh, NC.

[2] Diwekar, Urmila M. 2014. *Batch Processing: Modeling and Design*. Boca Raton: Crc Press, p. 41.

[3] "Relative Volatility - Encyclopedia Article - Citizendium." 2013. Citizendium.org. 2013. https://en.citizendium.org/wiki/Relative_volatility.

Appendices

Experimental Data

Table 1: Temperature readings, masses of empty container and sample, and refractive indexes of sample readings for trial 1

Time (mins)	**Temperature**		**Mass of Empty Container**		**Mass of Sample + Container**		**Mass of Sample**		**Refractive Index of Sample**		**Voltage**		**Watts**	
1	70.4	±	122.8	±0.05	133.1	±0.05	10.3	±0.05	1.342	±0.05	152	±	5.4	±0.05
2	70.5	±	121.6	±0.05	129.8	±0.05	8.2	±0.05	1.3408	±0.05	152	±	5.4	±0.05
3	70.5	±	120.7	±0.05	130.8	±0.05	10.1	±0.05	1.3407	±0.05	152	±	5.4	±0.05
4	70.5	±	122.3	±0.05	130.2	±0.05	7.9	±0.05	1.3407	±0.05	152	±	5.4	±0.05
5	70.5	±	120.9	±0.05	132.3	±0.05	11.4	±0.05	1.3408	±0.05	152	±	5.4	±0.05
6	70.6	±	121.5	±0.05	130.8	±0.05	9.3	±0.05	1.3409	±0.05	152	±	5.4	±0.05
7	70.6	±	121.2	±0.05	131.5	±0.05	10.3	±0.05	1.341	±0.05	152	±	5.4	±0.05
8	70.7	±	123.4	±0.05	134.7	±0.05	11.3	±0.05	1.3411	±0.05	152	±	5.4	±0.05
10	70.7	±	123.1	±0.05	145.2	±0.05	22.1	±0.05	1.3414	±0.05	152	±	5.4	±0.05
12	70.8	±	122.4	±0.05	143.3	±0.05	20.9	±0.05	1.3417	±0.05	152	±	5.4	±0.05
14	70.9	±	122.4	±0.05	144.5	±0.05	22.1	±0.05	1.3418	±0.05	152	±	5.4	±0.05
16	71	±	121.4	±0.05	144.4	±0.05	23	±0.05	1.3421	±0.05	152	±	5.4	±0.05
18	71.1	±	121.6	±0.05	146	±0.05	24.4	±0.05	1.3423	±0.05	152	±	5.4	±0.05
20	71.2	±	122.5	±0.05	147.1	±0.05	24.6	±0.05	1.3426	±0.05	152	±	5.4	±0.05
23	71.3	±	121.8	±0.05	156.8	±0.05	35	±0.05	1.343	±0.05	152	±	5.4	±0.05
26	71.5	±	121.4	±0.05	157.1	±0.05	35.7	±0.05	1.3434	±0.05	152	±	5.4	±0.05
29	71.6	±	121	±0.05	161.8	±0.05	40.8	±0.05	1.3438	±0.05	152	±	5.4	±0.05
32	71.8	±	123.2	±0.05	159.4	±0.05	36.2	±0.05	1.3443	±0.05	152	±	5.4	±0.05
35	71.9	±	121	±0.05	173.2	±0.05	52.2	±0.05	1.3447	±0.05	152	±	5.4	±0.05

Table 2: Temperature readings, masses of empty container and sample, and refractive indexes of sample readings for trial 2

Time (mins)	Temperature		Mass of Empty Container		Mass of Sample + Container		Mass of Sample		Refractive Index of Sample		Voltage		Watts	
1	70.7	±	122.8	±0.05	137.4	±0.05	14.6	±0.05	1.3433	±0.05	152	±	5.4	±0.05
2	70.7	±	121.6	±0.05	137.3	±0.05	15.7	±0.05	1.3432	±0.05	152	±	5.4	±0.05
3	70.8	±	120.7	±0.05	135.7	±0.05	15	±0.05	1.3435	±0.05	152	±	5.4	±0.05
4	70.8	±	122.3	±0.05	140	±0.05	17.7	±0.05	1.3438	±0.05	152	±	5.4	±0.05
5	70.9	±	120.9	±0.05	136.3	±0.05	15.4	±0.05	1.3441	±0.05	152	±	5.4	±0.05
6	70.9	±	121.5	±0.05	140.1	±0.05	18.6	±0.05	1.3442	±0.05	152	±	5.4	±0.05
7	71	±	121.2	±0.05	136.9	±0.05	15.7	±0.05	1.3445	±0.05	152	±	5.4	±0.05
8	71	±	123.4	±0.05	142.5	±0.05	19.1	±0.05	1.3446	±0.05	152	±	5.4	±0.05
10	71.1	±	123.1	±0.05	158.1	±0.05	35	±0.05	1.3449	±0.05	152	±	5.4	±0.05
12	71.2	±	122.4	±0.05	157.7	±0.05	35.3	±0.05	1.3452	±0.05	152	±	5.4	±0.05
14	71.3	±	122.4	±0.05	158.2	±0.05	35.8	±0.05	1.3457	±0.05	152	±	5.4	±0.05
16	71.5	±	121.4	±0.05	157.5	±0.05	36.1	±0.05	1.346	±0.05	152	±	5.4	±0.05
18	71.6	±	121.6	±0.05	157.9	±0.05	36.3	±0.05	1.3463	±0.05	152	±	5.4	±0.05
20	71.7	±	122.5	±0.05	159.2	±0.05	36.7	±0.05	1.3468	±0.05	152	±	5.4	±0.05
23	71.9	±	121.8	±0.05	179.3	±0.05	57.5	±0.05	1.3473	±0.05	152	±	5.4	±0.05
26	72	±	121.4	±0.05	177.5	±0.05	56.1	±0.05	1.3478	±0.05	152	±	5.4	±0.05
29	72.2	±	121	±0.05	177.6	±0.05	56.6	±0.05	1.3484	±0.05	152	±	5.4	±0.05
32	72.4	±	123.2	±0.05	179.6	±0.05	56.4	±0.05	1.349	±0.05	152	±	5.4	±0.05
35	72.6	±	121	±0.05	200.5	±0.05	79.5	±0.05	1.3499	±0.05	152	±	5.4	±0.05

Error Calculations

No error calculations were required, therefore there is no page for error analysis equations.

Raw Data Tables

Table 3. Raw Data Table for Trial 1

Time (min)	Temperature (°C)	Mass of Empty Container (g)	Mass of Sample + Container (g)	Mass of Sample (g)	Refractive Index of Sample	Voltage (V)	Watts (W)
1	70.4 ±	122.8 ±	133.1 ±	10.3 ±	1.3396 ±	152 ±	5. ±
2	70.5 ±	121.6 ±	129.8 ±	8.2 ±	±	±	±
3	70.5 ±	120.7 ±	130.8 ±	10.1 ±	±	±	±
4	70.5 ±	122.3 ±	130.2 ±	7.9 ±	±	±	±
5	70.5 ±	120.9 ±	132.3 ±	11.4 ±	±	±	±
6	70.6 ±	121.5 ±	130.8 ±	9.3 ±	±	±	±
7	70.6 ±	121.2 ±	131.5 ±	10.3 ±	±	±	±
8	70.7 ±	123.4 ±	134.7 ±	11.3 ±	±	±	±
10	70.7 ±	123.1 ±	143.2 ±	20.1 ±	±	±	±
12	70.8 ±	122.4 ±	143.3 ±	20.9 ±	±	±	±
14	70.9 ±	122.4 ±	144.5 ±	22.1 ±	±	±	±
16	71.0 ±	121.4 ±	144.4 ±	23 ±	±	±	±
18	71.1 ±	121.6 ±	146.0 ±	24.4 ±	±	±	±
20	71.2 ±	122.5 ±	147.1 ±	24.6 ±	±	±	±
23	71.3 ±	121.8 ±	156.9 ±	35.1 ±	±	±	±
26	71.5 ±	121.4 ±	157.1 ±	35.7 ±	±	±	±
29	71.6 ±	121.0 ±	161.8 ±	40.8 ±	±	±	±
32	71.8 ±	123.2 ±	159.4 ±	36.2 ±	±	±	±
35	71.9 ±	121.0 ±	173.2 ±	52.2 ±	±	±	±

Table 4. Raw Data Table for Trial 2

(mins)	Temperature	Mass of Empty Container	Mass of Sample + Container	Mass of Sample	Refractive Index of Sample	Voltage	Watts
1	70.7 ±	122.8 ±	137.4 ±	14.6 ±	±	±	±
2	70.7 ±	121.6 ±	137.3 ±	15.7 ±	±	±	±
3	70.8 ±	120.7 ±	135.7 ±	15 ±	±	±	±
4	70.8 ±	122.3 ±	140.0 ±	17.7 ±	±	±	±
5	70.9 ±	120.9 ±	136.3 ±	15.4 ±	±	±	±
6	70.9 ±	121.5 ±	140.1 ±	18.6 ±	±	±	±
7	71.0 ±	121.2 ±	136.9 ±	15.7 ±	±	±	±
8	71.0 ±	123.4 ±	142.5 ±	19.1 ±	±	±	±
10	71.1 ±	123.1 ±	158.1 ±	35 ±	±	±	±
12	71.2 ±	122.4 ±	157.7 ±	35.3 ±	±	±	±
14	71.3 ±	122.4 ±	158.2 ±	35.8 ±	±	±	±
16	71.5 ±	121.4 ±	157.5 ±	36.1 ±	±	±	±
18	71.6 ±	121.6 ±	157.9 ±	36.3 ±	±	±	±
20	71.7 ±	122.5 ±	159.2 ±	36.7 ±	±	±	±
23	71.9 ±	121.8 ±	179.3 ±	57.5 ±	±	±	±
26	72.0 ±	121.4 ±	177.5 ±	56.1 ±	±	±	±
29	72.2 ±	121.0 ±	177.6 ±	56.6 ±	±	±	±
32	72.4 ±	123.2 ±	179.6 ±	56.4 ±	±	±	±
35	72.6 ±	121.0 ±	200.5 ±	79.5 ±	±	±	±